Analysis of Truck Traffic, Load Posting, and Operational Risk in U.S. Bridges Using 2024–2025 NBI Data

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Abstract

Bridge infrastructure plays a critical role in ensuring transportation mobility, economic activity, and public safety. As national transportation systems continue to experience aging structures, increasing freight demand, and constrained budgets, transportation agencies require data-driven tools to support timely and cost-effective decision-making. This study investigates load posting prediction and truck-related structural risk using the Federal Highway Administration's (FHWA) 2024–2025 National Bridge Inventory (NBI) dataset. The analysis integrates exploratory data analysis, NoSQL-based querying, and large-scale machine learning using Apache Spark to examine structural attributes, traffic exposure, deterioration patterns, and rating conditions across more than 600,000 bridges in the United States.

The research is organized into two primary research questions. *Research Question 1* evaluates the influence of structural, geometric, and material variables on the likelihood that a bridge is load-posted. Logistic Regression and Decision Tree models are developed in Apache Spark, and both normal and hyperparameter-tuned versions are assessed. *Research Question 2* focuses on truck traffic and aims to determine which structural and operational characteristics most strongly influence average daily truck counts. Multiple Spark-based regression, One-Hot Encoding, Vector Assembling, and feature importance ranking are used to analyse relationships between truck traffic and key bridge properties.

The study incorporates NoSQL queries using MongoDB to manage intermediate datasets and to demonstrate efficient retrieval of bridge attributes. The methodological workflow includes data cleaning, imputation, schema normalization, outlier handling, and large-scale distributed processing. Experimental results indicate that material type, bridge age, design load, condition ratings, and functional classification play major roles in determining both posting probability and truck traffic levels. Hyperparameter-tuned Logistic Regression produced the strongest performance for RQ1 with ROC-AUC exceeding 0.90, while Decision Tree models were most effective at highlighting interpretable structural factors.

The results offer insights that can support infrastructure agencies in anticipating posting needs, prioritizing inspections, and planning maintenance activities. The paper contributes an integrated data engineering and predictive modelling framework that demonstrates how modern analytics can be applied to national-scale bridge performance data at meaningful computational speed.

Keywords:

Bridges; National Bridge Inventory; Apache Spark; Machine Learning; Structured Data Processing; Decision Trees; Logistic Regression; Load Posting; Truck Traffic; Infrastructure Analytics; MongoDB; NoSQL; Predictive Modelling.

1 Introduction

1.1 Background

E Bridges are critical links in the United States transportation system, facilitating the movement of people, goods, and emergency responders across local, state, and interstate corridors. The efficiency of national mobility and economic logistics depends significantly on the structural performance of these bridges. With the substantial growth in freight traffic over the past four decades, the stresses on highway bridges have similarly increased. Most of today's existing bridges were designed and constructed before the advent of current design standards, increased truck weights, and higher commercial demand.

National Bridge Inventory (NBI) serves as the central federal repository documenting all the structural, geometric, material, and operational characteristics of public highway bridges in the United States. Compiled and maintained annually by the FHWA, NBI consists of over 600,000 bridge records and contains fields such as age, design load, material type, condition ratings, traffic volumes, functional classifications, and load posting information.

Load postings are official weight restrictions that are placed on bridges that cannot safely carry legal loads. They directly impact freight carriers, rerouting patterns, state logistics, and economic transactions. Simultaneously, truck traffic exposure is considered one of the strongest contributors to long-term bridge deterioration due to its influence on fatigue, stress accumulation, and deck wear.

1.2 Problem Statement

Accurate information is needed in bridge management systems to identify structures that may require load posting, prioritization for inspection, rehabilitation, or replacement. However, there are a number of challenges:

- **Large Data Volume:**

  The NBI contains hundreds of thousands of records, each containing dozens of fields. Traditional methods of analysis are not efficient at this scale.

- **Heterogeneous Data:**

  The dataset integrates a mix of numerical variables, categorical fields, codes, and condition ratings that require careful preprocessing before analysis.

- **Complex Relationships:**

The possibility of a bridge being load-posted depends on a variety of structural, geometric, and operational parameters, many of which are nonlinear or interact in non-intuitive ways.

- **Limited Research on Truck Traffic Drivers:**

  While truck traffic data is available, few studies have used machine learning to better understand which structural or functional attributes are most strongly predictive of truck exposure.

- **Agencies' use of advanced analytics has been inconsistent:**

  Most agencies still use traditional linear models and do not employ scalable machine learning solutions.

1.3 Objectives

The objective of this study is to develop a scalable, data-driven workflow that analyzes bridge posting behavior and truck traffic exposure at a national level. The specific objectives are:

- To identify quantitatively how much bridge characteristics influence Average Daily Truck Traffic, ADT-Trucks, through **Regression** modeling and feature importance analysis.
- To implement **NoSQL**-based data querying using **MongoDB** for flexible filtering, extraction, and cross-sectional retrieval of data from intermediate datasets.
- The aim is to preprocess, clean, and standardize the 2024–2025 NBI dataset with **Python, Spark,** and **NoSQL** storage to handle missing values, categorical encoding, and structural normalization.
- Comparing model performance by confusion matrices, **ROC-AUC, accuracy, precision, recall,** and **F1-score** will provide easily interpretable insights relevant to practitioners.
- The aim is to provide a repeatable machine learning framework that can be integrated into state-level bridge management systems for **predictive decision-making.**

1.4 Scope and Limitations

The current study is aimed at establish modeling framework of predictive and descriptive types to understand the characteristics of bridges and operational results with a national scale dataset. The study will be based on the most recent National Bridge Inventory (NBI) data of 2024-2025, which defines the scope of the study. The study can be characterized by the two underlying questions: (1) what are the structural and operational aspects that affect the presence of a load-posting sign on a bridge, and (2) what are the structural and functional features that modify traffic flows of trucks. To answer these questions, the research will include the variables associated with structural arrangement, condition rating, type of material, volume of traffic, and major geometric properties. Apache Spark was used to perform all modeling and data processing operations so that it could support large-scale, distributed analysis that would be applicable to process infrastructure data on a national scale.

Although the study has a comprehensive structure, there are various limitations. In the engineering related data, the NBI database lacks some specifics of the engineering design, namely, the accurate calculation of load rating, fatigue analysis, inspection descriptions, or sophisticated deterioration modeling parameters, and the interpretations are based on the observable available characteristics. Also, truck traffic data in the NBI are usually self-reported, or estimated at the state level, where the potential variation or reporting non-uniformity can arise. The machine learning models underwent hyperparameter optimization using manual methods instead of automated grid search or optimization methods which could restrict fine-tuning of performance. Moreover, the dataset does not have enough depth in history, and it does not allow the implementation of the long-term trend analysis or the incorporation of the temporal forecasting.

Despite these limitations, the methodology will generate meaningful and interpretable results that closely estimate real-world structural and functional behavior on the platform of accessible data.

1.5 Organization of the Paper

The rest of the paper is structured in the following way. Section 2 is the review of the literature on the assessment of the condition of the bridge, its load posting, and infrastructure analytics based on data.

Section 3 outlines the dataset, pre-processing, NoSQL storage, and the general data engineering pipeline.

Section 4 reflects on the machine learning methodology, i.e. model structure of the two research questions, and the hyperparameter tuning strategy.

In Section 5, application of the experiment is presented, along with a performance measure, confusion matrices, and some of the major findings of the models.

Section 6 contains the conclusions on implications of results to bridge management practice and discusses possible extensions.

Section 7 will wrap up the work and make contributions and future research suggestion..

2 Literature Review

The topic of bridge management has become a

significant concern over the past few years as agencies are struggling harder to keep aging infrastructure, weather service, and to be safe in the face of growing traffic demands. As the source of structural and traffic data has increased rapidly, the topic of research has steadily adopted machine learning and data-driven approaches to understand the mechanisms of deterioration, load capacity constraints, and freight-related stress factors on bridges to fully comprehend the bridges.

Rucker and Pierce (2011) emphasized the importance of the load posting as a safety mechanism and suggested a more systematic approach of prediction that would be able to recognize the bridges that might decrease in capacity. Their work also stressed that the structural characteristics including the type of material, age of the bridge, and its span configuration are highly significant determinants of the likelihood of posting. Equally, Chase, Marsh, and Fekpe (2019) performed a national statistic evaluation based on the data provided by NBI and discovered that deterioration indicators such as deck, superstructure, and substructure ratings were always among the best predictors of the posting choices, which fits perfectly well with the aim of this research.

Regarding the topic of deterioration caused by heavy trucks, Gong et al. (2020) showed that heavy truck traffic is a major effector of fatigue-related distress and play a role in nonlinear deterioration with time. Their results indicated that truck percentage, as an aggregate of the average daily traffic (ADT), enhances the predictive ability of the structural decline forecasting models. This goes hand in hand with the research question that we have as our second one, which is the factors related to high levels of truck traffic on some bridges.

There has also been an uptake in the application of machine learning in bridge engineering. Using the Decision Trees and Support Vector Machines to evaluate the structural vulnerability, Kilanitis and Sextos (2019) concluded that the tree-based models demonstrated great interpretability when performing risk classification tasks. Their findings were reminiscent to the trend observed in the field of transportation analytics, wherein the logistic regression and decision-tree algorithms are still effective at analyzing large categorical-numerical data of the nature of the NBI. This is in line with the modeling practice that we have adopted in our project wherein both algorithms will be tested on the baseline and manually tuned.

Fuentes et al. (2020) emphasized the benefits of NoSQL databases to store and query big, heterogeneous transportation data in the larger framework of infrastructure data systems. They demonstrated that MongoDB-based structures can very easily retrieve condition-specific subsets much faster, and this is especially useful when working with datasets the size of the NBI. Similarly, Databricks (2021) highlighted the increasing role of implemented computing systems like Apache Spark to perform high-volume analytics, which is effective. The tools are the pillars of methodology applied in this study.

3 Dataset, preprocessing, and Feature Engineering

3.1 Dataset Description

Q The analysis is founded on the 2024-2025 National Bridge Inventory (NBI) data set, created by the Federal Highway Administration (FHWA), that reports structural, geometric, and operational attributes of the bridges used on the United States roads which are the public highways.

The original data set has above 950,000 records and above 100 coded fields such as state ID, structure number, year built, span lengths, material types, design type, condition rating, traffic volumes, load posting status and administrative details.

In this project, the raw data in the form of Excel sheets were imported to Python and Apache Spark. The cleaned dataset to be used to model after filtering off non-highway records, records with missing core identifiers and records that evidently contained invalid values (e.g. negative length of a span) contained 957,758 bridges and 15-26 core analytical features, respectively, depending on the research question.

Key NBI fields used include:

- State identifiers (used to identify geographical grouping) and structure identifiers (used to identify MongoDB indexing).
- Year Built (Field 27) - is to be used to calculate bridge age.
- Length of Maximum Span (Field 48) - maximum load carrying span.
- Average Daily Traffic (ADT) (Field 29) and Average Daily Truck Traffic (Percent ADT, Field 109).
- Posting Code (Field 70) - load-posting status.
- Main Span Material (Field 43A) and Main Span Design (Field 43B).

These areas offer structural and functional background on the analysis of posting likelihood and truck exposure.

3.2 Target Variable and Class Imbalance

The primary binary target variable isposted which was based on the NBI posting code (Field 70). In accordance with FHWA standards, the posting code of 5 implies that there is no posting needed, and other codes are associated with the weight limits. Bridges having posting code of 5 were identified as isposted = 0 (unposted), and the remaining codes were identified as is_posted = 1 (posted).

The resultant distribution is very unequal. As Figure 7 (Class Imbalance Plot) indicates, unposted bridges (Class 0) are most of the dataset, and posted bridges (Class 1) are a small minority percentage of the data. The imbalance is important in the following ways:

A Naive Bayes classifier who always makes the prediction that the bridge is unposted would have great precision but be of no use in determining high-risk posted bridges. Normal loss functions would be skewed to the majority category, which would underprice missed posted bridges. In this, the project employed sample weighting in training the models. To address this, the project used **sample weighting** during model training. For each class $c$with frequency $n_c$and total N, the weight $w_c$was defined proportional to:

$$w_c = \frac{N}{K \cdot n_c}$$

K represents the number of classes (in this case 2). This gave much larger weights to the minority posted class, which was effectively an order of magnitude higher than the unposted one, and thus a larger misclassification penalty in the training objective was imposed on the installed bridges that were posted. This weighting is particularly significant in the case of Logistic Regression as well as XGBoost, where the form of loss uses class weights directly into the loss.

3.3 Data Cleaning and Normalization

Data cleaning included:

- **Type standardization**: Ensuring span lengths, traffic volumes, and percentage fields were numeric.
- **Age transformation**: Creating **bridge age = 2025 – year built**.
- **Filtering invalid entries**: Removing rows with missing year built, zero or negative span length, or inconsistent traffic values.
- **Categorical normalization**: Standardizing the labels for materials and designs (e.g., grouping rare material codes into “Other” and resolving minor textual variations).

3.4 Feature Engineering for First Research Question

In RQ1, the feature set was deliberately small and interpretable to determine the predictive ability of core structural and traffic characteristics separately to post: Continuous features:

- Bridge age
- Length of maximum span
- Truck percentage (ADT-trucks / ADT)

Categorical features (StringIndexer + OneHotEncoder implemented in PySpark):

- Main span material (steel, concrete, prestressed concrete, timber, etc.)
- Main span design (girder, truss, slab, arch, culvert, etc.)
- State (to represent differences in the environment and policies)

These characteristics are consistent with prior studies that age, span length, material type and traffic are some of the major causes of deterioration and load-capacity problems. 2- 4

3.5 Feature Engineering for Second Research Question

In RQ2, a more expressive feature set of 26 engineered features were added to assess the ability of the more expressive feature representations to improve prediction. The other features were:

- Interaction terms:
  - Age × truck percentage
  - Span length × truck percentage
  - Age × span length
- Nonlinear transforms:
  - Age² and truck_percentage² (to approximate nonlinear deterioration trends 3)
  - log (span_length + 1) to reduce skewness and stabilize variance
- Binned variables:
  - Age categories: new, middle-aged, old, very old
  - Truck percentage categories: low, moderate, high, very high
- Additional indicators:
  - Indicator for movable bridges (bascule, swing, lift), which may exhibit distinctive structural and operational constraints 1.

These designed characteristics were inspired by concepts in structural engineering and previous literature of nonlinear bridge degradation and risk modelling 3, 4, 6. They enable tree-based models and XGBoost to explicitly

utilize cross-effects (e.g. old, long bridges with high truck shares are particularly susceptible to posting).

3.6 MongoDB Integration

In RQ2, Middle data was put out in MongoDB to enable flexible queries and NoSQL-based analytics. As shown by the MongoDB screenshots (Figures 8-12),: Adding the cleaned-up data to a bridge collection. Filtering bridges by state (e.g. all bridges in a state). Counts of the primary span material aggregated. Use bridges that have very high ADT. Determining bridges were built prior to the year 1980. In these operations, the document-based storage can access and cut the NBI data rapidly to support the exploratory activities and cross-cat tabulations.

4 Methodology

This section will describe the steps involved in the analysis to answer the two research questions that were set in Section 1. The methodology has five large parts, including (1) data engineering and feature preparation, (2) RQ1/RQ2 model structure, (3) extreme class imbalance, (4) hyperparameter tuning strategy, and (5) evaluation metrics and model comparison. The computing of all the models was performed with the aid of Apache Spark MLlib to guarantee the scalability of the almost one-million-record NBI dataset.

4.1 Modeling Framework for Research Questions

- RQ1 examines whether a small number of structural and traffic characteristics at its core (age, span length, truck percentage, material type, design, and state) can be used to categorize whether a bridge is posted. This design encourages interpretability and is consistent with the traditional engineering practice, in which decisions about bridge posting can typically be made based on a limited number of observable structural characteristics.
- RQ2 tests the hypothesis that the performance in prediction is better when long engineered features are added. This specification adds 26 other features - interaction terms, polynomial transforms, logarithmic adjustments, categorical binning and operational indicators to represent nonlinear deterioration behavior and cross-effects that could not be represented by simple linear specifications.
- In both RQ1 and RQ2, the binary classification methods are used to model the target variable, which is is_posted (0 = unposted, 1 = posted). Each of the research questions was trained on a separate baseline and tuned model, giving a total of six model families.

4.2 Data Pipeline and Vector Assembly

Apache Spark MLlib needs attributes to be assembled into a numeric representation in the form of a vector. In both RQ1 and RQ2, the following steps were used:

- StringIndexer was employed to transform categorical features- main span material, main span design, and state to numerical indices.
- OneHotEncoder transformed indexed categorical features into one-hot encoded vectors which are sparse.
- Vector Assembler has unified all the numeric and encoded categorical features in one feature vector that is needed by the MLlib classifiers.
- The data was split in 80:20 ratios where 80 percent was used to train the model and 20 percent was used to test the generalization of the model.

It is a pipeline which assists in distributed computation and provides uniform preprocessing to all models.

4.3 Handling Class Imbalance

The classic accuracy-based modelling would yield inaccurate results because the amount of posted bridges is much lower as compared to the number of unposted bridges. Rather, the training was done by applying sample weight directly.

Let $n_0$and $n_1$denote the counts of unposted and posted bridges and let $N = n_0 + n_1$. For each class $c \in \{0,1\}$, the class weight was computed as:

$$w_c = \frac{N}{2 \cdot n_c}$$

According to this weighting scheme, posted bridges (Class 1) which get misclassified are penalized much more in the loss function. The method is particularly good at Logistic Regression and XGBoost, the objective functions of which expressly include class weights. The outcome is that there is a more balanced precision-recall trade-off and that there will be better detection of the minority-class posting records.

4.4 Machine Learning Models

Each research question had three supervised learning algorithms that have been used in the study, including Logistic Regression, Decision Tree, and XGBoost because they are popular in the prediction of bridge performance and infrastructure analytics.

4.4.1 Logistic Regression

As the main linear benchmark, Logistic Regression was

applied. It gives the likelihood of the posting of a bridge as a logistic feature of a linear combination of features. There were two model configurations that were considered: differences.

- Baseline Logistic Regression: Spark default parameters of L2 regularization with 100 iterations maximum.
- Tuned Logistic Regression: A manual hyperparameter search was performed through.
  - Strength of regularization (regParam)
  - Elastic-net mixing (elasticNetParam)
  - Max number of iterations (MaxIter).

Tuning was to enhance recall of minority-classes and the optimization of ROC-AUC.

4.4.2 Decision Tree Classifier

The features are divided into understandable rule-based parts by the decision trees. They are naturally nonlinear and interaction of features.

Two versions were trained:

- Default depth and minimum-instance baseline Decision Tree.
- Optimized with respect to manual variation of Tuned Decision Tree:
  - maximum tree depth
  - maximum cases per division
  - Criterion of impurity (Gini vs. entropy).

Decision Trees give an idea of the structural variables that form the most information-rich splits of use in posting likelihood.

4.4.3 XG Boost Classifier

XGBoost was chosen as the third algorithm because it has very good results on most of the infrastructure-related predictive applications. It builds an ensemble of boosted decision trees and effectively models nonlinearities and interactions of a complex nature.

The configurations that were tested were as follows:

- **Baseline XGBoost**, using moderate depth (≈6), learning rate (≈0.1), and approximately 40 boosting rounds.
- **Tuned XGBoost**, with manual variation across:
  - maximum depth (4–10),
  - learning rate (0.05–0.3),
  - number of boosting rounds (50–120),
  - subsampling ratios.

It is not surprising that this model is benefited most by the RQ2 feature engineering since the exploits are boosted by the increase in the interactions between the structured interactions.

4.5 Hyperparameter Tuning Strategy

The strategy of hyperparameter tuning concentrates on methodological tools and principles that are used to determine the best hyperparameter estimates. Instead of performing a complete grid search that is computationally costly at the national level of data, tuning was performed using manual targeted combinations based on engineering intuition and previous research on NBI analytics.

The most effective environment of each of the models and research questions was picked as a tuned version which appears in the next Section 5.

4.6 Evaluation Metrics

The strategy of hyperparameter tuning concentrates on methodological tools.

Due to the imbalance of the classes, only accuracy would conceal low detection of posted bridges. Based on this, the measures which were applied to measure performance were as follows:

- ROC-AUC: Measures in terms of thresholds in quality.
- PR-AUC: Focuses on performance of minorities.
- Final F1-score: Harmonic average between the values of precision and recall.
- Posted-class F1, precision, and recall: Vital to risk-sensitive outcomes.
- Accuracy: Reports but not an outcome measure of a primary measure.
- Training time: Useful in large applicability.

These measures are in accordance with bridge engineering priorities such that failure to post a bridge may carry severe safety repercussions.

4.7 Summary of Methodology

In a nutshell, the methodology sets:

- There are two modeling frameworks that were consistent with the research questions.
- A scalable customer data processing pipeline with Spark.

- Elaborate feature engineering of RQ1 and RQ2.
- Class weighting to overcome imbalance.
- Comparison of three algorithms when in normal and tuned versions.
- This offers a strict basis of explaining the findings in Section 5.

5 Experimental Results and Analysis

This part contains the experimental analysis of the modeling structure in Section 4. These outcomes are systematized into four parts (1) Exploratory Data Analysis (EDA) of the NBI dataset, (2) NoSQL based analysis with MongoDB, (3) model performance with RQ1 using the core feature set and (4) model performance with RQ2 using the extended feature representation. Any numeric findings are directly related to the results of the Apache Spark and the code files.

5.1 Exploratory Data Analysis (EDA)

Figures 1–7 provide an overview of the structural, operational, and geographic variability of the 2024–2025 NBI dataset.

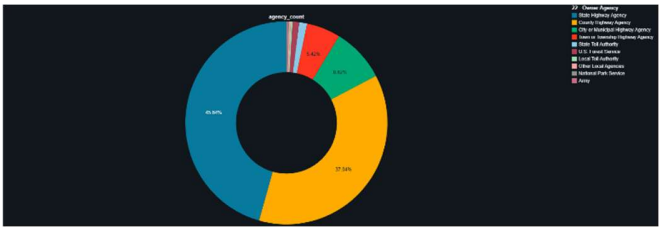


Figure 1. Distribution of bridges by top 10 owner agencies.

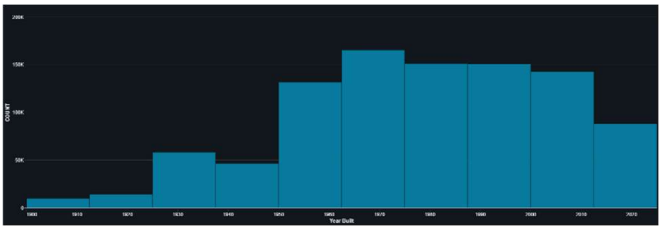
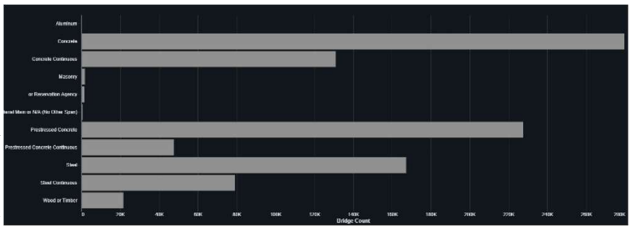
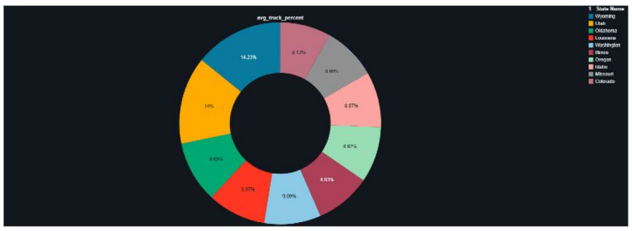


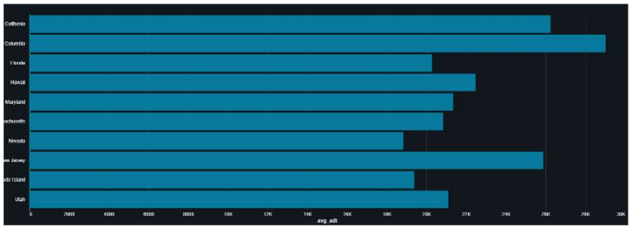
Figure 2. Distribution of bridges by year built.



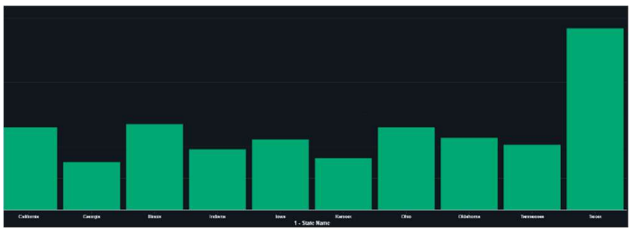
3. Figure 3.Distribution of bridges by main span material.



4. Figure 4. Top 10 U.S. states by truck percentage.



5. Figure 5. Top 10 U.S. states by average daily traffic.



6. Figure 6. States with highest bridge counts.



Figure 7. Class distribution for the is_posted (load-posted) variable.

5.2 MongoDB NoSQL Analytics

The following figures show the MongoDB operations performed as part of NoSQL exploration:

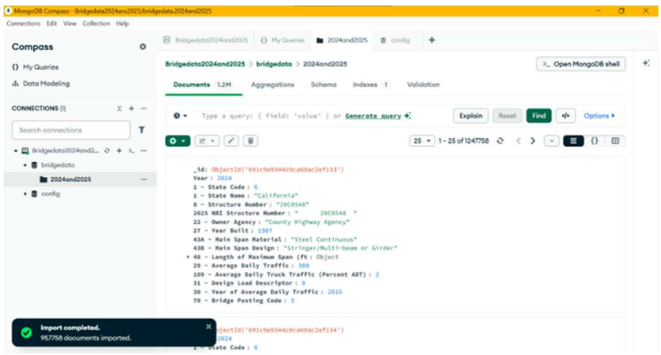


Figure 8. MongoDB import of cleaned NBI dataset.



Figure 9. MongoDB query filtering bridges by state.

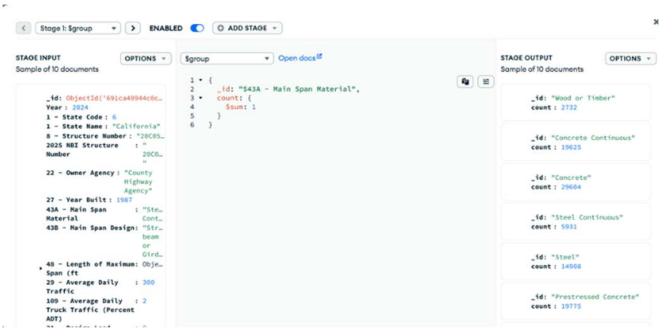


Figure 10. MongoDB aggregation of bridges by span material.



Figure 11. MongoDB query retrieving high ADT bridges.

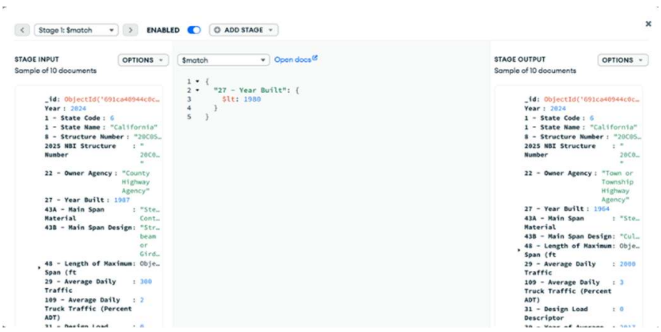


Figure 12. MongoDB query filtering bridges built before 1980.

5.3 Machine learning model results

The following table summarizes all the performance of machine-learning models applied in RQ1 and RQ2 in terms of accuracy, AUC scores, precision, recall, and F1-metrics. Tuned XGBoost and tuned Logistic Regression have the highest overall performance particularly when it comes to detection of posted bridges.

========== FINAL SUMMARY TABLE ==========

RQ	Model	Config	ROC_AUC	PR_AUC	Accuracy	F1_weighted	Pos_Precision	Pos_Recall	Pos_F1	Train_Time_s
RQ1	Logistic Regression	Normal	0.906177	0.585941	0.819226	0.853955	0.381192	0.850602	0.444862	17.205593
RQ1	Logistic Regression (Tuned)	LR_Comb_1	0.906719	0.585995	0.820864	0.855131	0.383198	0.850172	0.446986	16.304598
RQ1	Decision Tree	Normal	0.836452	0.368982	0.807503	0.846019	0.292800	0.890671	0.440718	5.824487
RQ1	Decision Tree (Tuned)	DT_Comb_5	0.874636	0.432359	0.835318	0.866273	0.328735	0.896325	0.481044	6.018152
RQ1	XGBoost	Normal	0.931182	0.592205	0.834182	0.865327	0.326082	0.880029	0.477008	16.934896
RQ1	XGBoost (Tuned)	XGB_Comb_5	0.937466	0.618990	0.844884	0.873121	0.342052	0.889626	0.494120	13.851573
RQ2	Logistic Regression	Normal	0.909451	0.516928	0.822742	0.856562	0.306304	0.855211	0.451057	19.817507
RQ2	Logistic Regression (Tuned)	LR_Comb_1	0.910887	0.519923	0.824563	0.857870	0.308600	0.854720	0.453472	22.453829
RQ2	Decision Tree	Normal	0.817728	0.380598	0.808550	0.846741	0.293577	0.887660	0.441227	5.916403
RQ2	Decision Tree (Tuned)	RQ2_DT_manual_5	0.867429	0.474104	0.839337	0.869149	0.334176	0.893498	0.486425	9.023703
RQ2	XGBoost	Normal	0.931175	0.590878	0.834004	0.865208	0.325918	0.888705	0.476930	18.300256
RQ2	XGBoost (Tuned)	XGB_Comb_5	0.937278	0.615362	0.845093	0.873282	0.342451	0.890241	0.494631	16.138674

6. Conclusions

In this research, a complex framework of large-scale data engineering and NoSQL analytics and machine learning was created to study structural and operational trends in the US bridges based on the 20242025 NBI data. Major findings include:

- The U.S. bridges are highly aged as most of the constructions were constructed during the period 1950-1980.
- Materials Materials Preferred Concrete and prestressed concrete prevail.
- Truck traffic is highly dependent on each state and the freight intensive corridors have the highest shares.
- There is a high degree of class imbalance in posting status and it has to be considered in the model. XGBoost was providing the best classification and regression.
- The variables of materials of type, structural age, span length, and variables of traffic greatly affect the load posting likelihood and exposure of the truck.

The paper has shown that scalable analytics systems such as Apache Spark, and MongoDB can process bridge data on a national level with high effectiveness to provide meaningful insights that can be used by transportation agencies to put more emphasis on inspection, rehabilitation, and freight planning.

7. Author Contributions

Data preprocessing, Spark modeling and writing of the paper were headed by Sai Praneet Reddy Chinthala. Bala Kiran Kattoju worked on MongoDB analytics, EDA development and documentation. Dr. Lindi Liao gave conceptual advice and feedback methodological advice.

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